NMIMS MPSTME | DESIGN EXPERIENCE

LONGITUDINAL MULTIMODAL PROGRESSION MODELING OF ORN

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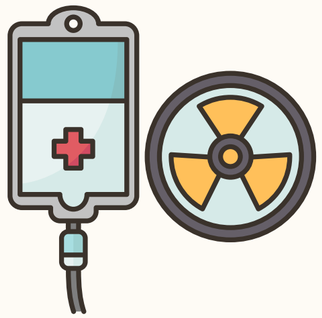
Overview

- Introduction
- Statement of the Problem
- Review of Related Literature
- Framework
- Hypothesis
- Methodology
- Scope and Limitations

What is Osteoradionecrosis?

Head and neck cancer patients receive Radiation Therapy to kill tumors

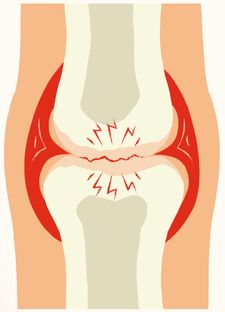


Radiation Therapy

Radiation damages the blood vessels that feed the jawbone

Blood vessels get damaged
Hypovascularity

Without proper blood supply, the bone slowly dies, leading to the problem



Bone loses blood supply

Radiation damages the blood supply to the jawbone, causing the bone to slowly die.

Bone dies (ORN)
Osteoradionecrosis

What is Osteoradionecrosis?

ORN is a severe, progressive late complication of radiation therapy for head and neck cancers — defined as non-healing, exposed bone within the irradiated field, in the absence of tumour recurrence.

The scale of the problem

Head and neck cancer affects 890,000+ people globally every year

ORN affects 3–15% of all HNC patients treated with radiation

Risk increases with chemotherapy, dental extractions, or radiation above 60 Gy.

The Clinical Challenge with ORN

The treatment ladder in its current state –

Conservative antibiotics + debridement	PENTOCLO protocol (pentoxifylline + tocopherol + clodronate)	Hyperbaric oxygen (HBO) therapy	Major resective surgery + free flap jaw reconstruction

WITH INCREASING SEVERITY →

The core problem – The right treatment depends entirely on the severity and trajectory of the disease — but current staging is subjective, inconsistent across specialties and blind to how the disease is evolving over time

Key evidence – A landmark 2025 study at MD Anderson tested 20 clinicians across oral oncology, radiation oncology, surgery, and neuroradiology — no specialty showed superior diagnostic accuracy, with significant disagreement between specialists.

HBO therapy is frequently prescribed yet only shows a ~38% success rate when used alone, and there is no validated model to predict which patients will respond.

The ongoing problem

Our current ML structure treats ORN as a binary, single-timepoint problem, question wether or not the person will develop ORN, this is computed over pre-treatment data.

This ignores biological reality. ORN is a dynamic, progressive disease that evolves across months to years post-radiation.

The four gaps this research addresses simultaneously

Gap	Current State	This Research
Temporal tracking	No model tracks ORN across time	Serial data across 5 timepoints
Multimodal integration	Imaging OR clinical OR dosimetric, never all	All four branches combined
Treatment output	No model recommends treatment	Treatment pathway recommendation
Patient population	Every dataset is Western (US/Europe)	South Asian cohort — first ever

Four Pillars Supporting This Research

Clinical Urgency	Clinically, the issue isn't whether a patient will get ORN—that's already known. The real question is: in Grade 2 ORN, will it progress, and should we operate now or wait? There's no validated tool to guide this.
Scientific Gap	Barua et al. is the only published longitudinal imaging paper in ORN — it involved just 21 patients, no biomarkers, and no ML. The West et al. (2026) dataset is longitudinal but has no imaging. The field is wide open.
Institutional Advantage	Tata Memorial's Patil et al. (2026) CT grading system, combined with its archive of 300+ HNC patients per year since 2015, creates a unique data asset no international group can replicate.
Journal Positioning	A validated longitudinal multimodal ML model with treatment triage is competitive for Lancet Digital Health (IF 23.8) or Nature Medicine — tiers unreachable with a binary risk predictor.

Approaches in the Literature

Study 1 — Humbert-Vidan et al. (2021), King's College London 96 patients. Compared logistic regression, SVM, random forest, AdaBoost, and neural networks using dose-volume histogram (DVH) parameters. Best result: ANN at 77% accuracy. Key finding: no statistically significant difference between any model, meaning the input features (DVH parameters alone) are the limiting factor, not the algorithm.

Study 2 — Reber et al. (2023), MD Anderson (n = 1,259) Largest direct comparison in the literature. Compared classical ML vs deep learning (ResNet, DenseNet) on full 3D dose maps. Result: Logistic regression (F1 = 0.3) outperformed all deep learning models, ResNet achieved F1 = 0.07. Conclusion: spatial dose data alone is insufficient. Additional imaging and non-dosimetric data must be added.

Study 3 — PREDMORN (Humbert-Vidan et al., 2025), 8 institutions, n = 3,928 Most rigorous multi-centre study to date. Stepwise logistic regression NTCP model, externally validated. Key predictors: mandibular D30%, dental extractions, smoking. Good discrimination, but still a static, pre-treatment binary classifier.

Other studies are covered in the thesis report

Study	N	Data Type	Timepoints	Key Limitation
Humbert-Vidan 2021 (London)	96	DVH + clinical	Single (pre-RT)	Tiny cohort, no imaging
Reber 2023 (MD Anderson)	1,259	3D CT dose maps	Single (pre-RT)	DL underperformed, no biomarkers
PREDMORN 2025 (8 institutions)	3,928	DVH + clinical	Single (pre-RT)	No imaging, no temporal data
West 2026 (MD Anderson)	1,129	Clinical + dosimetric	Multi (3–24mo)	No imaging, no biomarkers
Patil 2026 (Tata Memorial)	35	CT + follow-up	Multi	Small n, no ML
Barua (case series)	21	Serial CT	Multi (6–36mo)	Descriptive only, no ML
PROPOSED — Tata Memorial	80–150	Imaging + biomarkers + clinical + dosimetric	5 timepoints (T0–T4)	First South Asian cohort

What Every Dataset Looks Like vs. What We're Proposing

The Tata Memorial Cohort — A Data Asset No One Else Has

Built-in Longitudinal Infrastructure Tata's structured follow-up protocol means serial CT imaging is already embedded in standard care — creating a de facto longitudinal imaging archive that most institutions around the world simply don't have.

Different Cancer Profile Western datasets are dominated by HPV-positive oropharyngeal cancers. Tata's case mix includes high rates of buccal mucosa, tongue, and floor-of-mouth cancers — anatomically closer to the mandible, with different dose geometry and ORN risk distribution.

Tata Memorial's high patient volume combined with standardized radiotherapy protocols creates a uniquely consistent dataset where variations in outcomes more likely reflect true biological and anatomical differences rather than noise from heterogeneous treatment approaches.

Stage 1
Data Ingestion: Four Parallel Branches

Imaging	Dosimetric	Salivary/ Serum Biomarkers	Clinical Variables
Serial CT scans at T0 (pre-RT), T1 (3 months), T2 (6 months), T3 (12 months), T4 (24 months). Preprocessed by resampling to 1×1×1mm, HU windowing, and mandible auto-segmentation using nnU-Net.	3D radiation dose distribution maps from the RT planning system. Key values extracted: Dmax, D30%, D50%, mean dose, and dose to mandibular sub-volumes (alveolar vs. basilar bone).	IL-6, IL-8, TNF-α, MCP-1 measured at each follow-up visit via Luminex xMAP multiplex immunoassay. Serum CTX (bone resorption) and P1NP (bone formation) if available.	Dental health score, tobacco type (cigarette / bidi / smokeless / betel nut), diabetes, alcohol, chemotherapy, pre-RT extractions, cancer site, T-stage, N-stage, radiation dose schedule.

Stage 2
Extracting Meaning from Raw Data

Radiomics	Deep Learning (CNN)	From Biomarkers	From Dose Maps
1,645 features extracted per timepoint per mandibular sub-volume — texture, shape, density patterns. MFPCA (Multivariate Functional PCA) then encodes how these features change over time into compact components.	3D ResNet-18 or DenseNet-40 on CT mandible crops outputs a 512-D feature vector capturing complex spatial patterns beyond radiomics; optionally replaced with ViT embeddings.	Log-transformation (cytokines follow log-normal distributions). Missing timepoints filled using MissForest (non-parametric imputation). Standardized to z-scores per analyte.	DVH vectors passed through a 2-layer MLP encoder. Full 3D dose maps processed by pre-trained 3D DenseNet-40.

The Core AI: Temporal Fusion Transformer

What it does – *Takes all four feature streams across all 5 timepoints (T0–T4) and learns patterns that evolve over time — not just snapshots.*

Primary architecture — Temporal Fusion Transformer (TFT):

- Self-attention over the time axis: learns which timepoints matter most
- Variable selection network: identifies which data branch contributes most at each timepoint
- First published in 2021 — never before applied to oncological complication monitoring

Fallback architecture — LSTM with Late Fusion: Separate LSTM per branch across T0–T4. Hidden states concatenated at the final timestep. Simpler and preferred if patient numbers fall below 100.
Competing-Risk Survival Head (DeepHit / Fine-Gray model): Models two simultaneous competing events: time-to-ORN-progression and time-to-death. Outputs a cause-specific hazard function over time — not just a probability score.

Four Clinical Outputs

Output 1 — ORN Progression Risk Score Continuous 0–1 score per patient per timepoint. High score → expedite imaging review and specialist referral.

Output 2 — CT Grade Classification Automated multiclass output: Grade 0–4 per Patil et al. (2026) CT system. Replaces subjective clinician grading with a reproducible, automated classification.

Output 3 — Treatment Triage Recommendation 4-class output with confidence intervals:

- Conservative management
- PENTOCLO protocol
- Hyperbaric oxygen therapy
- Surgical resection

Output 4 — SHAP Explanation Card Per-patient report showing exactly which features drove the recommendation. Example: "IL-6 spike at T2 and mandibular basilar erosion on T3 CT most influenced the surgical recommendation."

Why Doctors Can Trust This Model — SHAP + Attention Maps

The problem with AI in medicine: Black-box predictions are clinically unusable. A doctor cannot act on a number without understanding why.

Two explainability tools used:

- **SHAP** (SHapley Additive exPlanations): Applied to the final dense layer. Ranks the contribution of every feature to every prediction. Shows which biomarkers, imaging features, and clinical variables drove the recommendation — and by how much.
- **TFT Temporal Attention Weights:** Visualized as heatmaps across timepoints and data branches. Shows the model when and what it was paying attention to. Example: high attention at T2 imaging + T3 IL-6 = the model flagged progression based on those two signals.

Clinical safeguard: Surgical resection (the highest-intensity recommendation) requires an additional clinician confirmation step. The model cannot trigger it alone.

Tools and Technologies

Component	Tool	Purpose
Segmentation	nnU-Net (PyTorch)	Auto-segment mandible from CT scans
Radiomics	PyRadiomics (Python)	Extract 1,645 texture/shape features per CT
Temporal Encoding	MFPCA (scikit-fda)	Encode time-series radiomics into components
Deep Learning	PyTorch + HuggingFace	3D ResNet/ViT embeddings; TFT architecture
Survival Modeling	DeepHit / lifelines	Competing-risk time-to-event modeling
Biomarker Imputation	MissForest	Non-parametric missing value imputation
Explainability	SHAP library	Feature attribution per prediction
Experiment Tracking	MLflow / Weights & Biases	Hyperparameter search, reproducibility
Clinical Output UI	Streamlit (Python)	Prototype clinician dashboard

Research Roadmap — Five Phases

- **Phase 1** | Months 1–2 — Systematic Review & Data Design Complete ORN ML literature review. Design data schema. Submit IRB/ethics application at Tata Memorial. Identify retrospective ORN-confirmed HNC cases (2015–2024). Pilot public TCIA dataset for architecture testing.
- **Phase 2** | Months 2–3 — Data Collection & Preprocessing DICOM de-identification and CT extraction. Mandible segmentation with nnU-Net + radiologist QA on 20% of cases. Grade every patient using Patil et al. (2026) CT system. Extract biomarker data from Tata biobank.
- **Phase 3** | Months 3–4 — Baseline Model Development Train XGBoost, Random Forest, LSTM, and single-timepoint CNN as baselines. Evaluate AUC, F1, calibration. Compare against PREDMORN NTCP benchmark.
- **Phase 4** | Months 4–5 — Temporal Fusion Model Implement TFT in PyTorch. Integrate all four branches. Add DeepHit competing-risk survival head. Hyperparameter tuning with Optuna, tracked via MLflow.
- **Phase 5** | Months 5–6 — Validation, Explainability & Write-up SHAP explanation cards. Temporal attention heatmaps. Compare against inter-clinician agreement benchmarks. Draft manuscript for Oral Oncology or Radiotherapy & Oncology. Prepare abstract for ASTRO/MASCC 2026.

Risks and How We Mitigate Them

Risk	Impact	Mitigation
Too few patients with serial imaging (target 100+)	High	Start case identification immediately; add prospective enrolment if n < 60; consider AIIMS Delhi collaboration
Biomarker data not available in existing records	High	Downgrade to optional module; core model uses imaging + clinical + dosimetric only
IRB/ethics delays at Tata Memorial	Medium	Submit in Month 1 alongside literature review; use retrospective waiver for de-identified data
Class imbalance (very few severe ORN cases)	Medium	SMOTE oversampling; class-weighted loss functions; calibrated probability outputs
Poor CT scan quality for older cases	Medium	Manual QA on first 50 cases; fine-tune nnU-Net for Tata's specific scanner
TFT underfits on small dataset	Medium	Pre-specify LSTM as fallback; transfer learning from TCIA HNC dataset
One semester not enough for full model	Low	Phase the deliverable: grade classification paper as primary output; TFT as ongoing research

Ethics, Fairness & Patient Safety

Patient Data Governance

- Full compliance with India's DPDP Act (2023) and ICMR Guidelines (2017)
- All DICOM data de-identified before analysis; dates shifted by randomized offset
- Data stored only on Tata Memorial institutional servers
- IRB ethics approval required before any data extraction

Model Fairness

- Treatment access bias: HBO is not uniformly available across India — historical treatment decisions will be documented alongside access constraints
- Tobacco type correlates with socioeconomic status — stratified analysis will ensure the model doesn't proxy deprivation as a clinical risk factor
- Any publication will explicitly state the model was trained on a South Asian cohort and may require recalibration before deployment elsewhere

Clinical Safety

- Positioned as decision-support only — not a replacement for clinical judgment
- All outputs include confidence intervals and SHAP explanation cards
- Surgical resection recommendation requires an additional clinician confirmation step — cannot be triggered by the model alone

How This Research Advances the Field

Dimension	Current Best	This Research
Temporal modeling	Single pre-treatment snapshot	5 timepoints (T0–T4) with TFT over time axis
Modality integration	Imaging OR clinical OR dosimetric	All four simultaneously
Clinical output	Binary ORN yes/no	Risk score + CT grade + treatment triage + SHAP card
Population	MD Anderson (US) or European only	First South Asian cohort — bidi, smokeless tobacco, oral cavity primaries
Survival modeling	Not applied to ORN ML	Competing-risk DeepHit: time-to-progression vs. time-to-death

What This Research Delivers

The problem in one line: ORN affects up to 1 in 7 HNC patients after radiation. Every existing AI model gives a static yes/no answer from pre-treatment data — and no treatment guidance.

What we are building: A multimodal, time-aware AI system that watches patients across 5 timepoints, combines imaging, biomarkers, radiation dose, and clinical history — and outputs a risk score, a CT grade, and a specific treatment recommendation with a plain-language explanation.

If successful, this model will:

Be the first longitudinal ML model for ORN anywhere in the world

Be the first to integrate salivary biomarkers with imaging in ORN

Be the first ORN AI tool built and validated on a South Asian population

Give clinicians a tool to answer the question they actually face: operate now, or wait?

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THANK YOU